

Federated-Boosting: A Distributed and Dynamic Boosting-Powered Cyber-Attack Detection Scheme for Security and Privacy of Consumer IoT

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Abstract—Consumer Internet of Things (CIoT) is an emerging technology that took the traditional consumer electronics to next level with smarter devices and higher connectivity. However, the rapid boom of this technology has captured a great deal of hacker's attention in the past decade. Due to resource restricted nature of CIoT devices and limited computational abilities, security professionals are unable to strengthen the security and privacy measures. Since the existing security methods suffers from data imbalance and overfitting issues, this paper proposes a distributed and dynamic weighted boosting method named Federated-Boosting for the accurate and timely detection of cyber-attacks in CIoT. Specifically, a weighting strategy is designed to overcome the data imbalance issue by dynamically adjusting the weights of weaker class samples. Additionally, a regularized loss function is designed that helps in controlling the overfitting of the model and provide enhance performance and generalization. Finally, a time-based and performance-based dynamic aggregation scheme is designed using distributed training model that keeps the privacy of the CIoT devices intact to the local server. Extensive experiments on two real-world datasets validates the superior performance of Federated-Boosting model against state-of-the-art detection models.

Index Terms—Security and privacy, cyber-attacks, consumer IoT, intrusion detection.

I. INTRODUCTION

A. Consumer Electronics

THE WORLD has witnessed significant advancements in the Consumer Electronics (CEs) products by adopting revolutionary technologies and tools, such as Internet-of-Things (IoT), cloud computing, Machine Learning (ML), etc. With this emergence and adaption, the U.S. Consumer Tech Industry reports that CEs revenues are expected to exceed

500 billion USD in the next few years [1]. Traditionally, CEs products are standalone which rarely connect to other devices or Internet. However, the next generation CEs market has been based products that have higher connectivity and enhanced intelligence as compared to traditional CE products. Hence, Consumer IoT emerged as a next level technology in the CEs field and studies [1] shows dramatic increase of IoT devices usage in different fields of CE products such as smart homes, healthcare, fintech, etc.

B. Consumer IoT in Healthcare

The urge to upgrade the conventional healthcare network (HCN) in to smart one is inevitable. Even though the concept of smart healthcare system has been talked about for decades, the existing HCN is faced with significant challenges, such as medical errors, lack of medical workers, and high cost [2]. Taking Europe and USA as example, where 10% and 20% of GDP is devoted to healthcare domain, respectively [3], which, yet, unable to avoid errors in medical decisions. In addition, it is projected that by 2030, the shortage of healthcare workers will reach to 18 million [3]. To overcome these challenges, the traditional CEs-based HCN are augmented and equipped with the IoT and Internet of Medical Things (IoMT)-based devices to enable enhanced communication characteristics and provide better healthcare services. This incorporation has revolutionized the typical CEs into the next-generation devices having sophisticated intelligence with enhanced connectivity measures. Thereby improving the automatic control and availability of data in the CE system through the connectivity of devices such as sensors, actuators, and other different consumer devices [2]. Nonetheless, these next generation CE devices are capable to be remotely accessed and controlled at any time, and from anywhere in the world.

C. Threat Significance

The market of CEs devices has remarkably progressed over the past decade. A study report shows that the user domain of CEs sector will reach up to 2, 873.1 million by 2025 with average revenue up to 317.10 billion USD [4]. Nowadays, each equipment or appliance may generate and communicate data online, leading to the expansion of CEs devices. The typical complex architecture of Internet connectivity faces challenges to acclimate to these modern-day

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Consumer Internet of Things (CIoT) devices and networks due to their dynamic nature. Thus, resulting in ineffective usage of resources, and ultimately exposing the network to diverse kinds of cyber-attacks [5]. Moreover, it is evident from the literature (please see Table I for example cyber-attacks) that CIoT-based smart systems are subject to numerous attacks, such as DoS, brute force, and botnets etc. [4].

D. Contribution

Federated learning (FL) concepts has emerged and shown advantageous provisions for delivering better security and privacy of intelligent medical systems [10], [11]. FL overcome the shortcomings of traditional centralized learning by offering distributed training models without accessing data of participating clients. Consequently, mitigating the privacy risks of sensitive user information and improves the learning process through shared computational resources, which might not be conceivable through centralized learning model. An intrusion detection system (IDS) could serve as the security backbone for the next-generation CIoT network, since it can easily detect the presence of malicious activities and attacks [8]. However, due to the specific requirements of CIoT networks (i.e., mobility, limited power and memory, and low) for real-time threat detection, the existing IDS faces significant design challenges. Additionally, the attack surface of threats is increasing day-by-day, which is why contemporary IDSs are incapable to identify unseen or *zero-day* attacks. Hence, a dynamic and adaptable security solution is needed for timely and accurate detection against evolving cyber-attacks in CIoT systems.

Therefore, this research work proposes a distributed boosting-based intrusion detection method for the security and privacy of CIoT. In summary, the proposed method improve the security domain in following ways:

- We design and develop an efficient Federated-Boosting model for security and privacy of CIoT networks that is capable of detecting *zero-day* attacks.
- We design an adaptive weighted boosting mechanism to accurately identify complex attack vectors that assign dynamic weights to overcome data imbalance issue.
- We design loss function regularization to eliminate the issue of over-fitting.
- We design a distributed IDS model using dynamic aggregation strategy to preserve the privacy of participating CIoT clients.

The remaining sections of this work are organized in following manner. The related work and research gap is presented in Section II. The details of proposed design is outlined in III. The experimental evaluations and results are presented in Section IV, while conclusion is presented in Section V.

II. RELATED WORK

A. Literature Review

The literature has reported a plethora of research work contributed towards the security of CIoT networks against a disseminated range of insider and outsider threats [12], [13], [14], [15]. For example, the authors of [16] developed a

security model deep learning-based method to detect common security threats like DoS¹ and port-based attacks. The authors reported high accuracy rates, however their proposed model is computationally expensive to be applied in restricted IoT networks. Similarly, the authors of [17] designed a Stacked Deep learning model that uses bio-inspired Spider Monkey optimization method. The model is capable of detecting various kinds of attacks such as DoS, probe, R2L² and U2R³ attacks with high accuracy. However, they evaluate their model with an outdated (NSL-KDD) dataset that does not represent the complex attack vectors in the contemporary IoT framework.

To detect DDoS⁴ attacks in IoT environments, the writers of [18] developed a benchmarking of several ML classifiers. Although they used both new and old datasets to evaluate their model, but they neglected the applicability of their model in real-world IoT networks. The authors of [19] developed a blockchain-based deep model for the security and privacy of IoMT networks. Specifically, they used autoencoder for the privacy and LSTM for security purpose. However, the use of blockchain technology makes it not feasible for CIoT-based healthcare networks because of obvious drawbacks such as high consumption and lack of clarity regarding the ownership of data. Similarly, the authors of [20] also developed a threat detection model using blockchain and deep learning for the security of industrial IoT networks. However, they did not consider the limitations of IoT devices in their developed model.

In view of federated learning [21], the literature also reported distributed learning methods for security of medical networks. For example, the authors of [22] developed a method named *SecFedNIDS* using federated learning to guard IoT against poisoning attacks. However, their method required a lot of human effort in the application stage. Similarly, the authors of [23] also proposed federated learning-based model for security enhancement of IoMT networks. In an effort to enhance privacy of the IoMT systems, the authors of [24] developed a model named *FL-BETS* to preserve the privacy and detect data frauds at local fog level nodes and remote cloud nodes. However, the detection of unknown attacks were not considered during the development phase which makes it inapplicable to detect *zero-day* attacks.

B. Research Gap and Motivation

It is evident from the literature review that most of the existing methods focuses on security aspect and neglected the privacy issue. Similarly, the performance and access control are the key driven factors of existing methods; hence, they lack deploy ability in CIoT environments. Additionally, the threat of maleficent clients in the aggregation mechanism posed a significant danger to the security and privacy of critical CIoT networks. Also, a tailored security mechanism is required to cope with the challenges raised by the heterogeneous devices

¹Denial-of-Service.

²Remote-to-Local.

³User-to-Root.

⁴Distributed Denial-of-Service.

TABLE I
CYBER-ATTACK INCIDENTS REPORTED IN DIFFERENT COUNTRIES

S.No	Country	Attack type	Result	Target	Reference
1.	USA	Malware	Cause permanent damage to components	Hancock regional hospital	[6]
2.	UK	Ransomware	Caused disruption in providing urgent healthcare services	NHS 111 service	[7]
3.	Germany	Ransomware	Data loss	Lukaskrankenhaus Neuss	[8]
4.	France	Ransomware	Data breach of up to 510,000 people	L'Assurance Maladie	[9]
5.	Norway	Data breach	Records of nearly 2.9m people had been compromised	Regional health authority	[6]

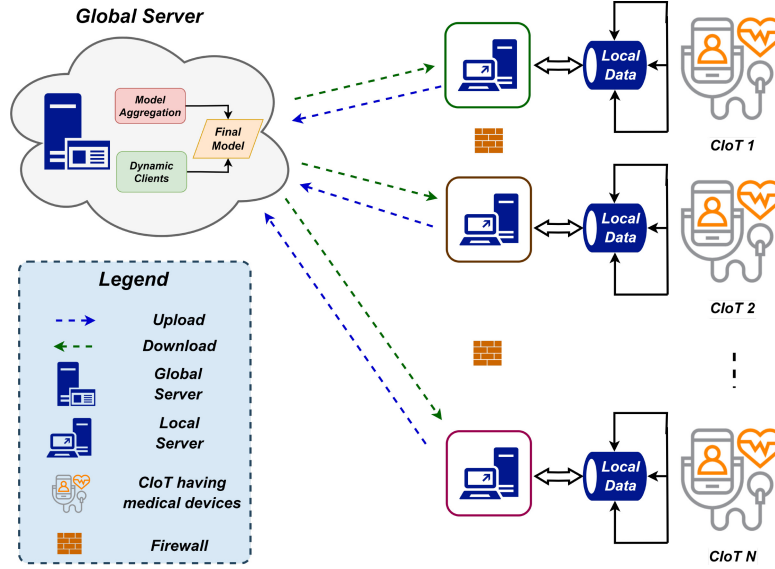


Fig. 1. General overview of proposed Federated-Boosting IDS model for CIIoTs.

in the modern CIIoT. To address these gaps, this paper put forward an end-to-end security design that does not only offer robust security and privacy scheme but also provide a dynamic aggregation system for malicious clients in the distributed architecture of CIIoT environments. To the best of our knowledge, there is no such study that offer cyber-attack detection mechanism for security and privacy of CIIoT using federated architecture.

We applied federated learning because of its distinct benefits such as: it allows the model training with multiple data sources while protecting data privacy; it decreases the need for large-scale data transfers between clients and servers, which is essential in bandwidth-constrained environments for example IoT/CIIoT; it doesn't incur a large computation overhead compared to training a model in a conventional way which increases performance factor; and It offers scalability by handling large number of devices or entities.

III. PROPOSED FEDERATED-BOOSTING DESIGN

A. Model Overview

In traditional or central networks, only the local state of a network can be perceived through central learning. Thus, it is challenging to create an overall design of a distributed structure and even harder for an IDS method to design a

powerful detection/defensive system against evolving cyber-attacks. On the other hand, federated architecture provides a global perspective of the entire network with unified controlling abilities, making distributed network flow obtainability easier. As shown in Figure 1, the proposed model has a combination of central global server having several local servers where each local server represents a distinct CIIoT network. The global server manages the weight and parameter updates, dynamic client list, aggregation mechanism and distribution of global models. The proposed model is placed on the global server for significant reasons such as: i) changing functionality and improved adaptability. ii) flexibility between the linked local servers and CIIoT devices. and iii) governing capability to control the entire network as a sole an intelligent decision engine. This global view of entire distributed network and devices provides a combination of security and privacy and thus develop an efficient cyber-attack detection system to defend against malicious actions. The Federated-Boosting model is flexible to fit more devices by simply updating the parameters to the global server.

B. Preprocessing

Preprocessing is an important stage in Machine Learning as the data quality and the extracted meaningful information can directly affects the capability of a learning model; hence,

it is very important to preprocess the input data before it is feed to the detection model. The Federated-Boosting model comprises the following preprocessing techniques:

1) *Feature Reduction*: This step is applied since CIoT devices generates high-dimensional and heterogeneous data. The aim of this application is to remove noisy and less relevant features from the input data as it can effectively helps in reducing the size and computational cost of the detection model. Since we intend to design a lightweight model for restricted CIoT network, it is vital to reduce the feature set to achieve several advantages like faster training time, low memory consumption, and less computational requirements. The Federated-Boosting model incorporates ICA⁵ technique to reduce the feature set, which is defined below by:

$$id = \mathcal{N}c + m \quad (1)$$

where the k -dimensional feature of the input data denoted by id , the m -dimensional independent components are denoted by c , the matrix for mixing constant $k \times m$ with $k \geq m$ and m shows noise, is denoted by the $\mathcal{N}$. Suppose the medical data is generated and labeled in noise-free environment, then the ICA method is shown below by:

$$id = \mathcal{N}c \quad (2)$$

and

$$c = \mathcal{U}id \quad (3)$$

where the matrix for un-mixing is represented by $\mathcal{U}$, to project id to c .

These independently shown components are reflected as the best feature space as it can provide higher estimates $\mathcal{N}$ and c having id with non-Gaussian data (which is standard practice in existing CIoT networks) optimizing constraint. Ultimately, the optimizing constraint of non-Gaussian data to decrease the shared information ($\mathcal{SI}$) between m variables (c_j , where $j = 1, 2, \dots, m$), is represented below to solve Eq. (4).

$$\mathcal{SI}(c_1, c_2, v_3, \dots, c_k) = \sum_j \mathcal{E}(c_j) - \mathcal{E}(c_e) \quad (4)$$

where $\mathcal{E}$ shows the differential entropy. We chose ICA over other methods because it can support to isolate mixed signals into their original independent components, making it easier to comprehend the underlying dynamics of data. Additionally, this method is selected because the CIoT devices generates non-Gaussian structure data and the devices and networks are assumed to be independent.

2) *Feature Normalization*: Since the proposed model is dependent on trees and weights, the high range of feature values can induce a bias in the learning system which eventually may cause some feature weights to update faster as compared to other features. Hence, it is important to address this issue by utilizing a statistical normalization method such as, the *Z-score* technique, as given below:

$$\mathcal{ZS}^{(j)} = \frac{id^{(j)} - \mu}{\sigma} \quad (5)$$

where σ denotes the standard deviation, and the mean values of m for a particular feature ($id^{(j)} (j \in 1, 2, \dots, m)$) is shown by μ .

C. Weighted Boosting Method

Boosting is an ensemble training strategy in which the outcomes of several training models (*aka* base models) is integrated to predict a cyber threat. In this work, two modified functions are applied to the boosting strategy, i.e., weight function (WF) and regularized learning objective or regularized loss function RLF). The WF helps in rectification of gradient deviation problem while the RLF term assists in avoiding the over-fitting issue by smoothing the final learnt weights and assist in the process of detecting *zero-day* attacks. This is done through the addition of penalty/regularization term to the usual loss function that demoralizes complex training methods having heavy weights. This is achieved through the regularization parameter in the training process in order to discourage the model for being too complex and therefore avoiding over-fitting issue. Additionally, this method enhances the generalization ability of the proposed model when executed on test data.

This weighting strategy effectively treat the data imbalanced problem by appropriately assigning large weights to minority or weak labels to bring out the best of classification results. Similarly, the over-fitting issue is addressed through the RLF strategy that tries to reduce the complexity of the model unlike other algorithms such as KNN,⁶ RF,⁷ NN,⁸ *etc.* However, to strongly learn every feature the model comprises of a large number of regression trees since the training data is skewed in nature. Let N denotes the tree count, and M denotes the records in training set, then the prediction outcome of p_m and N^{th} tree for input data record is given by:

$$p_m^N = p_m^{N-1} + nt_N(b_m) \quad t_n \in \mathcal{T} \quad (6)$$

where p_m^{N-1} represents the outcome of previous tree, which is added to $t_n(b_m)$ that represents the newly constructed tree and produce the p_m^N prediction result, m denotes the learning rate, and $\mathcal{T}$ represents the collection of probable trees. Here we can optimize the objective function to minimize the prediction error as shown below:

$$\mathcal{OF}^N = \sum_{m=1}^M \mathcal{L}(x_m, p_m^N) + \gamma tr_c + \frac{1}{2} \lambda we^2 \quad (7)$$

where $\mathcal{OF}^N$ represents the objective function, the correct value for m^{th} sample class is denoted by x_m , λ and γ represents the parameters of regularization that are employed to control the tree pruning and complexity, respectively, tr_c denotes the number of terminal nodes, and we represents weight of each leaf in tree. The boosting model utilized in this work try to optimize $\mathcal{OF}^N$ via second-order estimates for the gradients of loss function to accurately learn about attack-related samples. However, the proposed method could estimate wrong gradients

⁶k-nearest neighbors.

⁷Random forest.

⁸Neural network.

⁵Independent Component Analysis.

if the intrusions are inserted by hacker/invaser at the rare nodes resulting in imbalance issue. Therefore, a WF is designed that allocates the $\mathcal{CW}$ class-weights to the corresponding classes, as shown by:

$$\mathcal{CW} = 2 \times \left(\frac{\mathcal{R}_m}{\mathcal{R}_c} \right) \quad (8)$$

where $\mathcal{R}_c$ denotes the total number of training records and $\mathcal{R}_m$ denotes each class records. In this was the training error is minimized by assigning large weight to minority class and less weight to majority class. Then the updated $\mathcal{OF}^N$ with $\mathcal{CW}$ is computed as:

$$\mathcal{OF}^N = \sum_{m=1}^M \left(\mathcal{CW}_m \mathcal{L} \left(x_m, \left(p_m^{N-1} + t_n(b_m) \right) \right) \right) + \gamma tr_c + \frac{1}{2} \lambda cw^2 \quad (9)$$

Then, to further optimize the $\mathcal{OF}^N$ the second-order Taylor estimation is calculated as:

$$\mathcal{OF}^N = \sum_{m=1}^M \left(\mathcal{CW}_m \left[gr_m t_n(b_m) + \frac{1}{2} h_m t_n^2(b_m) \right] \right) + \gamma tr_c + \frac{1}{2} \lambda cw^2 \quad (10)$$

where h denotes the hessian (h_m) = $\frac{\delta^2 \mathcal{L}(x_m, p_m^{N-1})}{\delta^2 p_m}$, and gr denotes the gradients $gr_m = \frac{\delta \mathcal{L}(x_m, p_m^{N-1})}{\delta p_m}$. Using these equations an efficient and strong tree splitting strategy can be applied for enhanced training of each feature and $\mathcal{CW}$ weights allocated to each link. Initially, since every tree is a leaf, therefore $\mathcal{T} = t(b) = \mathcal{CW}_{s(b)}$ denotes a leaf-node record. If the records of this leaf-node is denoted as $J_i = r | s(b_r) = i$ via substituting to Equation (10), we attain the following:

$$\mathcal{OF}^N \cong \sum_{i=1}^{tr} \left(\mathcal{CW}_m \left[\mathcal{G}_i cw_i + \frac{1}{2} (\mathcal{H}_i + \lambda) cw_i^2 \right] \right) + \gamma tr_c \quad (11)$$

where $\mathcal{H}_i = \sum_{r \in J_i} h_m$ and $\mathcal{G}_i = \sum_{r \in J_i} gr_m$. Suppose a fixed size tree structure, then the optimum weight value can be calculated via $\mathcal{OF}^N$ to 0, and the cw value denoted as $cw = -\frac{\mathcal{G}_i}{\mathcal{H}_i + \lambda}$. To explore all appropriate splits, alternatively we employed greedy algorithm to pick the most valuable split in each tree for the generation of weighted quantiles which are most applied in multi-trees splitting. Initially for each tree, the gain ($\mathcal{G}$) is calculated, then a quantile is selected based on maximum value of $\mathcal{G}$ since it represent the best split. Ultimately, the value of $\mathcal{G}$ is calculated for every tree using its right (Rb) and left (Lb) branch as below:

$$\mathcal{G} = \frac{1}{2} \left[\frac{\mathcal{G}_{Lb}^2}{\mathcal{H}_{Lb} + \lambda} + \frac{\mathcal{G}_{Rb}^2}{\mathcal{H}_{Rb} + \lambda} - \frac{(\mathcal{G}_{Lb} + \mathcal{G}_{Rb})^2}{\mathcal{H}_{Lb} + \mathcal{H}_{Rb} + \lambda} \right] - \gamma \quad (12)$$

Throughout the training step this whole process is repeated for all existing sub-trees until all the hyper parameters and $\mathcal{OF}^N$ are optimized to develop an effective IDS model that precisely distinguish between normal and malicious labels.

D. Dynamic Federation and Aggregation Strategy

The distributed and collaborative client-based architecture of federated learning design makes it perfect fit for use in CIoT/IoMT networks since they contain private and sensitive data. The learning process is done both at the local client side and global server side. Each client then shares their model updates with the server through an aggregation mechanism (for example, federated averaging) after which the server generates the global model and distribute it to the local clients. This aggregation and updation process is repeated through iterations/communication rounds until a threshold performance is achieved. A brief explanation of this federated process is summarized as follows:

- Initialize and distribute the global model (if exist) as $\mathcal{GM}_I^0$.
- If it does not exist, then initialize the training process at the server and distribute the global model as $\mathcal{GM}_I^{cr}$.
- The local server receives the global model $\mathcal{GM}_I^{cr}$ and swap it with the local model (if exist) denoted as $\mathcal{LM}_n^{cr}$ where n represents local server/client and cr represents the communication round. The local server trains its own model using its local data and tries to optimize via reducing the loss function represented as $\mathcal{LL}(\mathcal{LM}_n^{cr})$. The optimized weights and parameters are sent as updates to the global server for aggregation purposes. The loss is calculated as:

$$\mathcal{LL}(\mathcal{LM}_n^{cr}) = \arg \min_{\mathcal{LM}_n^{cr}} \mathcal{LL}(\mathcal{LM}_n^{cr}) \quad (13)$$

$$\mathcal{LL}(\mathcal{LM}_n^{cr}) = \frac{1}{k_n} \sum_{i=1}^{k_n} (t_i - f(r_i))^2 \quad (14)$$

where the number of training records is represented by k_n , the specific training record is shown by r_i and its label is shown as t_i , and the final output of $\mathcal{LM}_n^{cr}$ model is shown by $f(r_i)$.

- The global server generates a list of dynamic participating clients for aggregation procedure. This dynamic aggregation client list is dependent on two factors; accuracy and time. The model results of only those local servers/clients will be aggregated which meets the aggregation strategy (such as achieved threshold accuracy performance and reply with updated parameter weights in certain amount of time). After receiving the updates, the global server generates the global model as $\mathcal{GM}_I^{cr+1}$ by training and loss minimization and optimization. This (global loss $\mathcal{GL}(\mathcal{GM}_I^{cr})$) process is shown as:

$$\mathcal{GL}(\mathcal{GM}_I^{cr}) = \frac{1}{K} \sum_j^K \mathcal{LL}(\mathcal{LM}_n^{cr}) \quad (15)$$

- The global server update the global model $\mathcal{GM}_I^{(cr+1)}$ from the participating client lists as:

$$\mathcal{GM}_{I(m)}^{(cr+1)} = \mathcal{GM}_{I(m)}^{cr} + \left[\frac{1}{\sum_{\Delta \mathcal{LM}_{n(m)}^{cr} \neq 0} |\mathcal{DS}_n|} \sum_{n=1}^s |\mathcal{DS}_n| \Delta \mathcal{LM}_{n(m)}^{cr} \right] \quad (16)$$

TABLE II
DETAILS OF EXPERIMENTAL SETTINGS

Equipment Details	Configuration
Operating system	<i>Ubuntu 18.04.2 LTS</i>
CPU	<i>Intel(R) Xeon(R) Silver 4110 2.10GHz</i>
Libraries	<i>Keras, TensorFlow, Scikit – learn, Numpy</i>
Environment	<i>Python</i>
GPU	<i>NVIDIA RTX2080 GPU</i>
Ram	<i>128Gb</i>

where the total number of dataset records at client n is represented by $|\mathcal{DS}_n|$, the total participating clients is denoted by s , and the count of non zero gradients transferred by the clients is denoted by $\sum_{\Delta \mathcal{LM}_{n(m)}^{cr} \neq 0} |\mathcal{DS}_n|$.

All these above-mentioned steps are executed through communication rounds until the global model converge and attain desirable accuracy.

IV. EVALUATION AND RESULTS

A. Experimental Settings

The performance evaluation of Federated-Boosting model is done through extensive experiments conducted on a workstation (details of which are specified in Table II). The booster is set as *gbtree*, the importance_type is set as *gain*, the learning rate is 1, with maximum depth is set as 10, with max_delta_step as 0, subsample ratio of the training instances is set as 0.5, refresh_leaf as 1, num_parallel_tree is 1, and sampling_method is set as *gradient_based*.

B. Datasets

The proposed model is primarily evaluated using two datasets from different CIoT domain to assess its performance and generalizability. The basic motivation for applying these datasets are: They are created from real-world CIoT-based networks; and They realistically represent medical data and the operations of consumer IoT (since there is a deficiency of realistic CIoMT datasets to evaluate machine learning algorithms for cyber security purposes).

1) *EHMS-2020 Dataset*: The EHMS⁹ dataset was generated through a real-time medical system testbed [25] by collecting both patient's biometrics and the network flow data. The total number of features in this dataset are 44 out of which 8 are related to biometrics of patients and 35 are related to network flow. Data were collected for different CIoT sensors such as Electrocardiogram, Blood Oxygen Saturation, temperature, and blood pressure sensors. The total number of records in this dataset are 16,318 out of which 14,272 (87.5%) are normal records and 2,046 (12.5%) are abnormal records and hence this ratio represents imbalance nature of this dataset. The abnormal records were generated using various attack vectors such as data injection, spoofing, and man-in-the-middle attacks.

⁹Enhanced Healthcare Monitoring System.

TABLE III
NUMERICAL RESULTS OF FEDERATED-BOOSTING METHOD
USING BOTH DATASETS

Evaluation Metric	EHMS (%)	ToN_IoT (%)
Precision	99.749	99.563
F-Score	99.741	99.546
Recall	99.516	99.156
AUC/ROC	99.742	99.549
Accuracy	99.745	99.554

2) *ToN_IoT Dataset*: This dataset was generated through mimic testbed architecture of IoT and IIoT systems [26]. This dataset contains data from heterogeneous types of CIoT devices (such as Fridge, GPS, Motion Light, Thermostat, etc.) to collect telemetry data of IoT services, network traffic and different attack scenarios were implemented such as DoS, Ransomware, and Backdoor etc. This dataset consists of 22 features and 156,119 abnormal records and 245,000 normal records.

C. Detection Results

This work employed well known metrics for the performance evaluation of proposed model such as Precision, F-Score, Recall, and Accuracy. The experiments were conducted by dividing the dataset into training and testing sets where the training data contained 80% records and testing set contained 20% records. The numerical results of the experiments are presented in Table III, which shows that the Federated-Boosting model achieved 99.749% Precision, 99.741% F-Score, 99.516% Recall, 99.742% AUC/ROC and 99.745% Accuracy for the EHMS dataset and 99.563% Precision, 99.546% F-Score, 99.156% Recall, 99.549% AUC/ROC and 99.554% Accuracy for the ToN_IoT dataset. These results also indicates the learning ability and robustness of the proposed model against *zero-day* attacks by proficiently identifying numerous types of complex attack vectors in an imbalanced datasets. Even though the ToN_IoT dataset contain a large number of complex cyber-attacks, the proposed boosting model is able to learn the underlying patterns in the CIoT data by successfully distinguishing them.

Furthermore, we conducted additional experiments using CIC-IDS-2018 [4] and WUSTL-IIOT-2021 [12] datasets to assess the performance of proposed model via traditional CEs data. It is worth mentioning that our model managed to achieved 99.291% and 99.434% accuracy for CIC-IDS-2018 and WUSTL-IIOT-2021 datasets, respectively.

D. Comparative Analysis and Discussion

We conducted extensive experiments (in terms of accuracy metric) to compare the performance of Federated-Boosting model with the SOTA methods.¹⁰ The parameter settings of these methods are taken from their respective papers and the

¹⁰State-of-the-art.

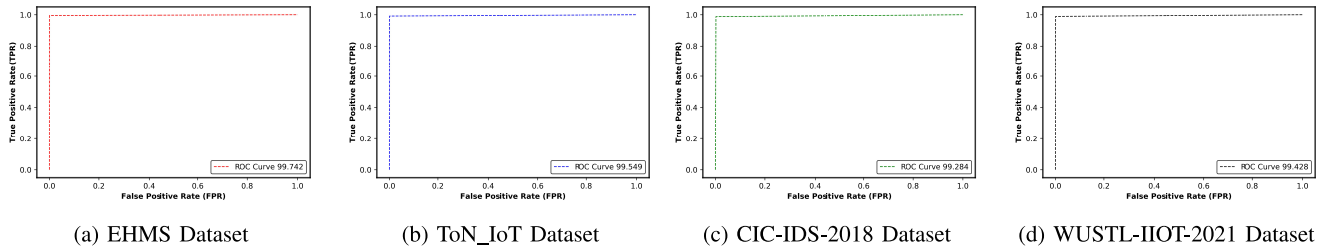


Fig. 2. Comparison of AUC performance on all four datasets.

TABLE IV
COMPARISON OF SOTA AND FEDERATED-BOOSTING METHOD USING
ACCURACY METRIC ON BOTH DATASETS

Comparison Method	EHMS (%)	ToN_IoT (%)
Grammatikis <i>et al.</i> [27]	92.102	90.326
Kumar <i>et al.</i> [28]	93.808	92.249
Samuel <i>et al.</i> [29]	95.221	94.582
Manimurugan <i>et al.</i> [30]	94.122	94.373
Hossen <i>et al.</i> [23]	95.025	95.446
Federated-Boosting	99.741	99.546

numerical outcomes are shown in Table IV. It is observed from this comparison that the Federated-Boosting model outperformed all the other detection methods by achieving high accuracy rate using both datasets. The proposed Federated-Boosting model shows an average improvement of 5.686% and 6.151% as compare to other methods on EHMS dataset and ToN_IoT dataset, respectively. These results reflect the advantage of dynamic aggregation strategy design to adapt to the dynamic behaviors of the contemporary CIoT networks. The use of weighted boosting design also significantly improves the detection performance as compare to other SOTA methods. The reason for enhanced performance of proposed model is that it is based on dynamic ensembles and boosting mechanism which assists in dealing and smoothing with the “bias-variance” trade-off. The proposed model has the potential to fit in small-scale platforms such as CIoT because of light-weight architecture and less computational requirements.

To summarize, the Federated-Boosting model is a lightweight and reliable security system for CIoT platforms. However, it entails a high volume data to achieve desired performance. In future, we plan to integrate few-shot training mechanism in the proposed model with incremental learning approach to overcome this limitation.

V. CONCLUSION

Existing cyber-attacks detection and defensive methods are proven to be ineffective against the complex and evolving nature of threats. This paper proposes an effective intrusion detection method for efficient detection of cyber-attacks in CIoT networks. The proposed model is developed using dynamic and adaptive boosting design in a privacy preserved manner by keeping intact the local data on the client side.

The aggregation mechanism is designed by dynamically selecting contributing clients in the federation strategy to deal with malicious clients. The proposed model is tested with two publically available datasets and proved to have good performance (accuracy = 99.74% for EHMS dataset, 99.54% for ToN_IoT dataset) using imbalanced data. This security design can serve as a starting point for security professionals and critical organizations to secure their infrastructure against cyber-attacks.

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